AFSHUR THOUFEEQ MUSHARRAF K

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SUMMARY

Master's graduate in Electrical Engineering (ASU, CGPA 3.75/4.0) specialized in Digital VLSI Design, with a strong foundation in digital electronics and CMOS and hands-on exposure to the RTL-to-GDSII flow: Verilog RTL, logic synthesis, floorplanning, CTS, routing, static timing analysis and physical verification, through ICC2-based physical design training and ASIC projects. Actively building hands-on skills in Synopsys and Cadence tools. Seeking entry-level ASIC roles in Physical Design, STA, Synthesis, and RTL/Logic Design.

EDUCATION

M.S. Electrical Engineering — Arizona State University, Tempe, Arizona, USA | CGPA: 3.75/4.0 May 2025
Relevant Coursework: VLSI Design, Digital Systems and Circuits, Digital Verification and Testing, Machine Learning Basics with Deployment to FPGAs, Analog IC Design, Microelectronics Manufacturing.

B.E. Electrical and Electronics Engineering — PSG College of Technology, Coimbatore, India | CGPA: 9.1/10 May 2022
Relevant Coursework: Digital Electronics, Computer Architecture, Analog Circuits, Microprocessors & Microcontrollers, Digital Signal Processing, Data Structures and Algorithms.

TECHNICAL SKILLS

Physical Design: Floorplanning, Power Planning, UPF Multi-Voltage Design, Placement, CTS, Routing, Timing Closure
STA & Sign-off: Static Timing Analysis (setup/hold, MCMM, DMSA), SDC Constraints, Parasitic Extraction (SPEF/PEX), Crosstalk, DRC, LVS
Digital Design & Synthesis: Digital Electronics, Verilog RTL, FSM Design (Mealy/Moore), Counters, Logic Synthesis (Design Compiler), CMOS, Low-Power Design
EDA Tools (training and academic projects): Synopsys (ICC2, Design Compiler, PrimeTime, StarRC, IC Validator, HSPICE), Cadence (Innovus, Virtuoso), Siemens Calibre, ModelSim
Scripting & Languages: Tcl, Linux, Python, Verilog, VHDL, C

PROFESSIONAL TRAINING / CERTIFICATION

ASIC Physical Design Trainee — VLSIGuru, Bangalore, India June 2026 – Present

- Completed foundation training in digital electronics, CMOS, low-power design, ASIC/PD flow, STA, Linux and Tcl scripting, followed by an advanced **Synopsys ICC2** physical design course.
- Implemented the complete netlist-to-GDSII flow in **Synopsys ICC2** on ORCA_TOP (28nm, SAED32 library, 435 MHz, 52K cells, 40 macros, 0.5 mm² core): floorplanning, power planning, placement, CTS and routing.
- Built a **UPF-based multi-voltage design** with 2 power domains (PD_ORCA_TOP, PD_RISC_CORE; 0.75V/0.95V). Wrote Tcl to create voltage areas from target utilization with guard band and verify them.
- Automated with **Tcl**: area and utilization reports, metal-layer rules (min width, spacing, pitch), HVT/LVT cell mix analysis (40.2%/59.8%) and port placement using pin constraints.
- Performed macro placement, congestion analysis, endcap/tap cell insertion and power grid creation. Fixed power-planning issues and max-transition violations through driver upsizing and buffer insertion.
- Achieved timing closure using CTS (NDR, useful skew) and setup/hold fixes across MCMM scenarios. Resolved post-route crosstalk, DRC and LVS issues, extracted SPEF with **StarRC** and ran STA sign-off in **PrimeTime** (DMSA).

ACADEMIC PROJECTS

ASIC Acceleration for Graph Convolutional Network (GCN) | *Verilog, Synopsys DC, Cadence Innovus, ModelSim, Calibre, 7nm ASAP7*

- Designed a low-power GCN accelerator in Verilog for node classification, using sparse **COO format**, FSM-controlled transformation and aggregation blocks, and **argmax** output.
- Reduced computation and memory by running transformation first and replacing adjacency matrix multiplication with COO-driven row additions.
- Verified against golden outputs in **ModelSim** at RTL, post-synthesis and post-APR stages. Synthesized with **Synopsys Design Compiler**.
- Completed RTL-to-GDSII in **Cadence Innovus** (floorplan, power planning, CTS, routing) at **1 GHz**: 0.178 mm², ~254K gates, 789.6 mW. Fixed DRC/connectivity violations and ran DRC/LVS checks in **Calibre**.

RTL-to-GDSII Flow for a 2-Bit Adder | *Verilog, ModelSim, Synopsys DC, Cadence Innovus, Calibre, 7nm ASAP7*

- Designed a 2-bit adder in Verilog with registered outputs and verified it in **ModelSim** at the RTL, post-synthesis and post-APR stages using one testbench.
- Synthesized with **Synopsys Design Compiler** using SDC constraints (500 ps clock): 11 cells, 21.9 µm², ~6 µW, with +435 ps setup slack.
- Completed **Cadence Innovus** APR (floorplan, power stripes, tap/filler cells, CTS, routing; 46.7 µm² core) and met timing after CTS (+440 ps setup, +2.6 ps hold)
- Exported GDSII to Virtuoso and ran Calibre DRC/LVS, with LVS matching the layout to the source.

Design, Layout and Verification of a NAND2 Standard Cell | *Cadence Virtuoso, HSPICE, Calibre, 7nm ASAP7 FinFET*

- Designed a NAND2xp5 standard cell (3-fin NMOS, 2-fin PMOS) in 7nm ASAP7 FinFET PDK, matching the 7.5-track cell library architecture.
- Simulated in HSPICE (TT corner, 0.7 V, 70°C) with an FO4 load and 25 ps input slews. Verified all four input combinations in WaveViewer and measured rise/fall times.
- Drew the custom layout in **Cadence Virtuoso**, passed **Calibre DRC/LVS** (3×3 array showed only the expected latch-up flag)
- Extracted parasitics with **Calibre PEX** and re-simulated post-layout (35.6 ps rise, 29.4 ps fall), analyzing how layout parasitics slow the cell.

Design and Custom Layout of a 4-Bit Full Adder | *Cadence Virtuoso, Synopsys IC Validator, HSPICE, 32nm CMOS*

- Designed a 4-bit ripple-carry adder from a custom mirror-adder cell in 32nm CMOS at 1.05 V (30 nm length, 300/450 nm NMOS/PMOS widths).
- Drew the custom 1-bit and 4-bit layouts in **Cadence Virtuoso** and passed **Synopsys IC Validator** DRC and LVS (clean).
- Simulated the post-layout netlist in **HSPICE** with a 4-inverter output load: **109 ps** worst-case delay, **57 µW** average power, **33.3 µm²** area.
- Beat all three specs (400 ps, 100 µW, 150 µm²) with an overall score of 9.9, against 3.0 for a design that only meets the specs.

WORK EXPERIENCE

Control Engineering Intern — Danfoss Drives, Loves Park, Illinois, USA June 2025 – Dec 2025

- Automated HIL Testing pipelines using Robot Framework, Typhoon HIL and MATLAB/Simulink.
- Optimized CI/CD performance via Git to reduce test execution times by 50%.

Process Automation Engineer — Dow, Chennai, Tamil Nadu, India June 2022 – July 2023

- Programmed DCS-based automated process controls using ABB800xA.
- Worked on Software migration, validation, and cross-functional hardware troubleshooting.